Digital System Design Lab

Mini Project

Servo control using potentiometer input

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Processes

ADC

The Nexys 4DDR has a built-in ADC which can be used to read the potentiometer wiper signal.

PWM Generation

Using the onboard 100 MHz clock input, a PWM signal of appropriate width and period is output to a pin.

Servo

The microcontroller inside the servo measures the received PWM signal and controls the motor as desired.

Challenges deep-dive

ADC Configuration

Using the IP catalog in Vivado, the XADC package is to be added to the project.

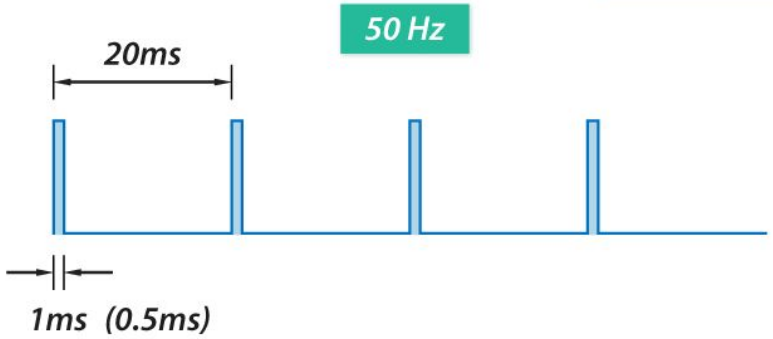
Timing calculation

After the analog value has been converted, mathematical operations are performed to calculate a time interval where the PWM output is HIGH.

Electrical Circuit

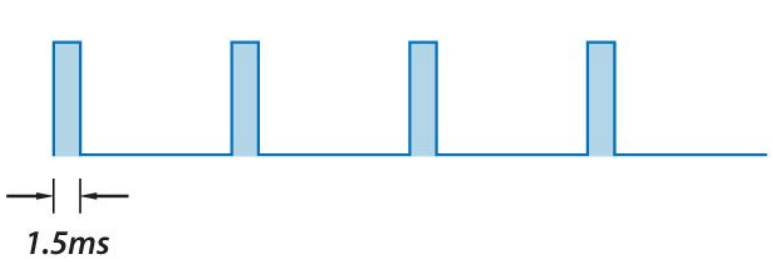
ADC input is connected through a voltage divider to limit to 1 Volt. Other ADC channels are grounded. FPGA PWM output pin is connected to servo PWM inputs.

SERVO MOTOR CONTROL



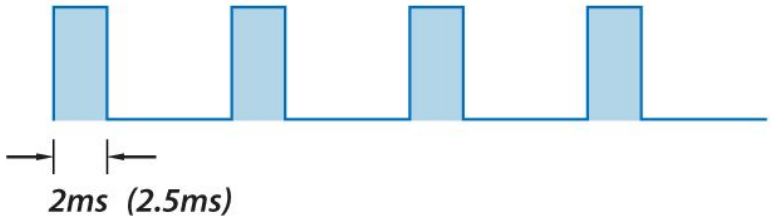
0 Degrees





90 Degrees





180 Degrees

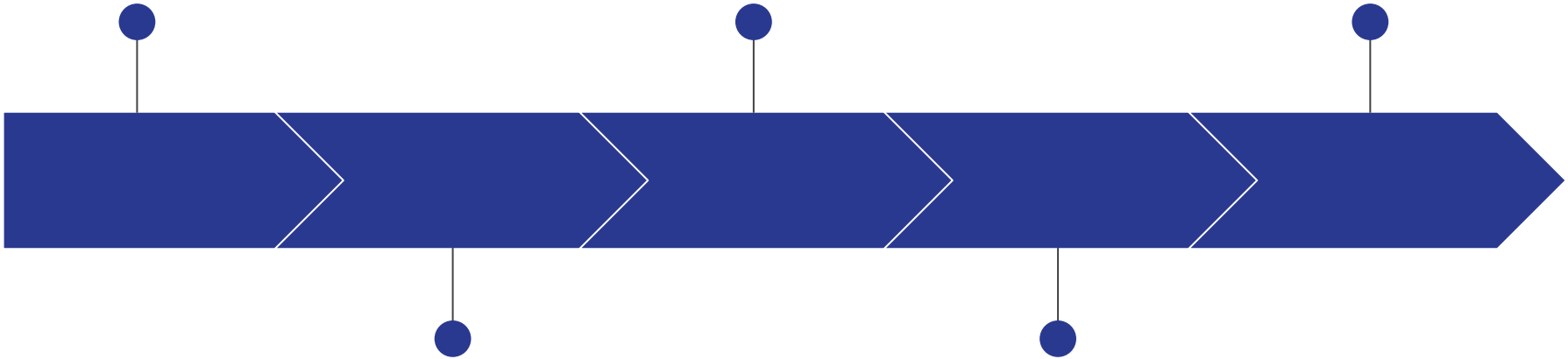


Implementation

Convert analog to digital => DUTY

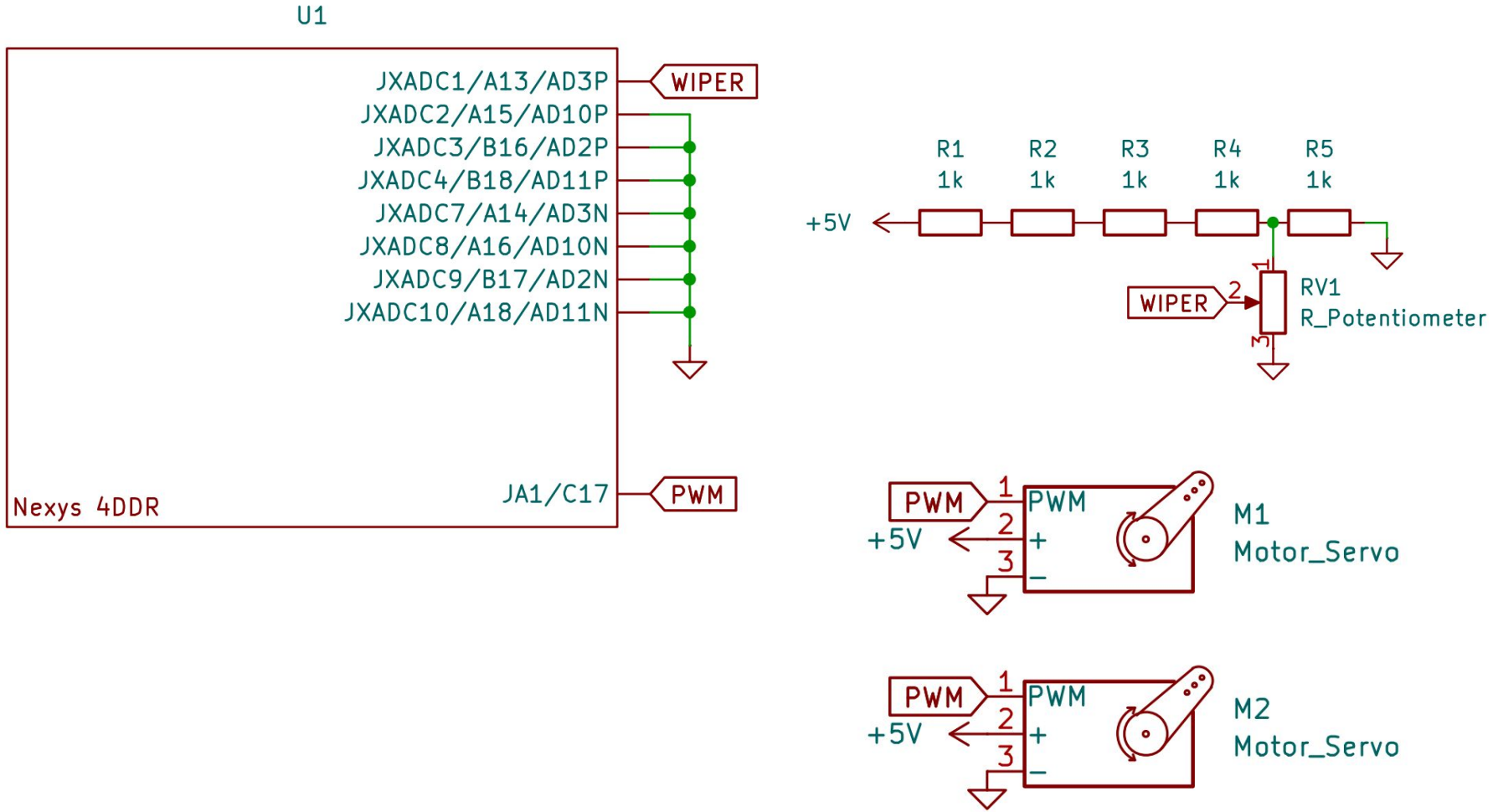
If count < interval => PWM HIGH

If count == 2e6 => repeat



Calculate interval => 500 us + DUTY * 1900 us

If count > interval => PWM LOW


U1
JXADC1/A13/AD3P
JXADC2/A15/AD10P
JXADC3/B16/AD2P
JXADC4/B18/AD11P
JXADC7/A14/AD3N
JXADC8/A16/AD10N
JXADC9/B17/AD2N
JXADC10/A18/AD11N
WIPER
JA1/C17
PWM
Nexys 4DDR
R1
1k
R2
1k
R3
1k
R4
1k
R5
1k
+5V
WIPER
RV1
R_Potentiometer
PWM
+5V
M1
Motor_Servo
PWM
+5V
M2
Motor_Servo